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Advanced SQL NULL Functions Exercise 2.1

Q1: SELECT employee_id,
name,
IFNULL (Salary, 0) AS salary-with-default
FROM Employees;

employee_id	name	Salary-with-default
1	Alice	5000
2	Bob	0
3	Charlie	7000
4	Dana	0

Q2: SELECT employee_id,
name,
IFNULL (department, 'Not assigned') AS department_name
FROM Employees;

employee_id	name	department_name
1	Alice	HR
2	Bob	IT
3	Charlie	Not assigned
4	Dana	Finance

Q3: SELECT order_id,
customer_id
FROM Orders
WHERE customer_id IS NULL;

order_id	customer_id
103	NULL

Q4: SELECT ^{order_id,}
customer_id,
FROM ~~del~~ orders
WHERE

Q4: SELECT Order_id,
customer_id
IFNULL (delivery_date, 'Pending') AS delivery_status
FROM orders;

Order_id	customer_id	delivery_status
101	201	2024-12-01
102	202	Pending
103	NULL	2024-12-03

Q5: SELECT student_id,
Name
IFNULL (grade, 0) AS final_grade
FROM students;

Student_id	Name	Final_grade
1	Ethan	85
2	Maya	0
3	Olivia	90

Q6: SELECT ~~IFNULL (grade~~ grade AS not_graded
FROM students
WHERE grade IS NULL;

not_graded
NULL

Q7: SELECT produce_id,
 name,
 IFNULL (discount, 0) AS final_price
 FROM Products;

Product_id	name	final_price
S01	keyboard	0
S02	Mouse	5
S03	Monitor	0

Q8: SELECT COUNT (email) AS missing_email_count
 FROM customers
 WHERE email ISNULL;

Missing_email_count
2

Q9: SELECT customer_id,
 name,
 IFNULL (email, 'No Email') AS email_display
 FROM Customers;

customer_id	name	email_display
1	Linda	No Email
2	Joseph	joseph@mail.com
3	Nia	No Email

Q10: SELECT payment_id,
 ~~Method~~, Status, ...
 IFNULL (Method, 'Unknown') AS Method_display,
 ~~FROM payments~~ Status
 FROM Payments;

Q10: Table output

Payment-id	Method display	Status
301	Credit	NULL
302	PayPal	Success
303	Unknown	Failed

Q11: SELECT Item_id,
Item_name,
IFNULL (quantity, 0) AS quantity_checked
FROM Inventory;

Item_id	Item_name	quantity_checked
1	Pen	0
2	Notebook	150
3	Eraser	0

Q12: SELECT emp_id
COALESCE (bonus, commission) AS first_available_reward
FROM Employee_Extra;

emp_id	first_available_reward
1	300
2	100

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Q13: ~~SELECT room AS~~
SELECT COUNT (room) AS no_room_count
FROM classes
WHERE classes ISNULL;

no_room_count
2

Q14: SELECT student_id,
date,
IFNULL(status, 'Not Marked') AS attendance_status
FROM Attendance;

student_id	date	attendance_status
1	2025-04-01	Not Marked
2	2025-04-01	Present
3	2025-04-01	Absent

Q15: SELECT account_id,
IFNULL(account_type, 'Unknown') AS type_display,
IFNULL(balance, 0) AS balanced_checked
FROM bank_accounts;

account_id	type_display	balanced_checked
A1	Savings	0
A2	Current	5000
A3	Unknown	2000

Q16: SELECT project_id, title,
IFNULL(start_date, 'TBD') AS start_display
FROM Projects;

project_id	title	start_display
1	Website Revamp	2025-01-10
2	Mobile App	TBD
3	Data Migration	TBD

Q17: SELECT review-id,
 product-id,
 IFNULL (comment, 'No comment') AS comment-display,
 IFNULL (rating, 0) AS rating-display
 FROM Reviews;

review-id	product-id	comment-display	rating-display
1	S01	Great product	4
2	S02	No comment	0
3	S03	Works fine	3

Q18: SELECT Supplier-id,
 name,
 COALESCE (phone, alt-phone, 'NO contact') AS Contact-number
 FROM Suppliers;

Supplier-id	name	Contact number
1	Global Goods	123 456789
2	Best Supplies	987654321
3	Value Source	No contact

Q19: SELECT user-id,
 IFNULL (theme, 'Light') AS theme-set,
 IFNULL (language, 'English') AS language-set,
 IFNULL (timezone, 'UTC') AS timezone-set
 FROM User-settings;

user-id	theme-set	language-set	timezone-set
1	Light	English	UTC
2	Dark	English	UTC+1
3	Light	English	UTC

Q20: Select record_id,
Machine_id,
IFNULL(issue, 'Unknown issue') AS issue_log,
IFNULL(technician, 'Not Assigned') AS technician_name
FROM Maintenance;

record_id	Machine_id	issue_log	technician_name
1	M01	Overheating	Not Assigned
2	M102	Unknown issue	Not Assigned
3	M103	Jammed	Alex